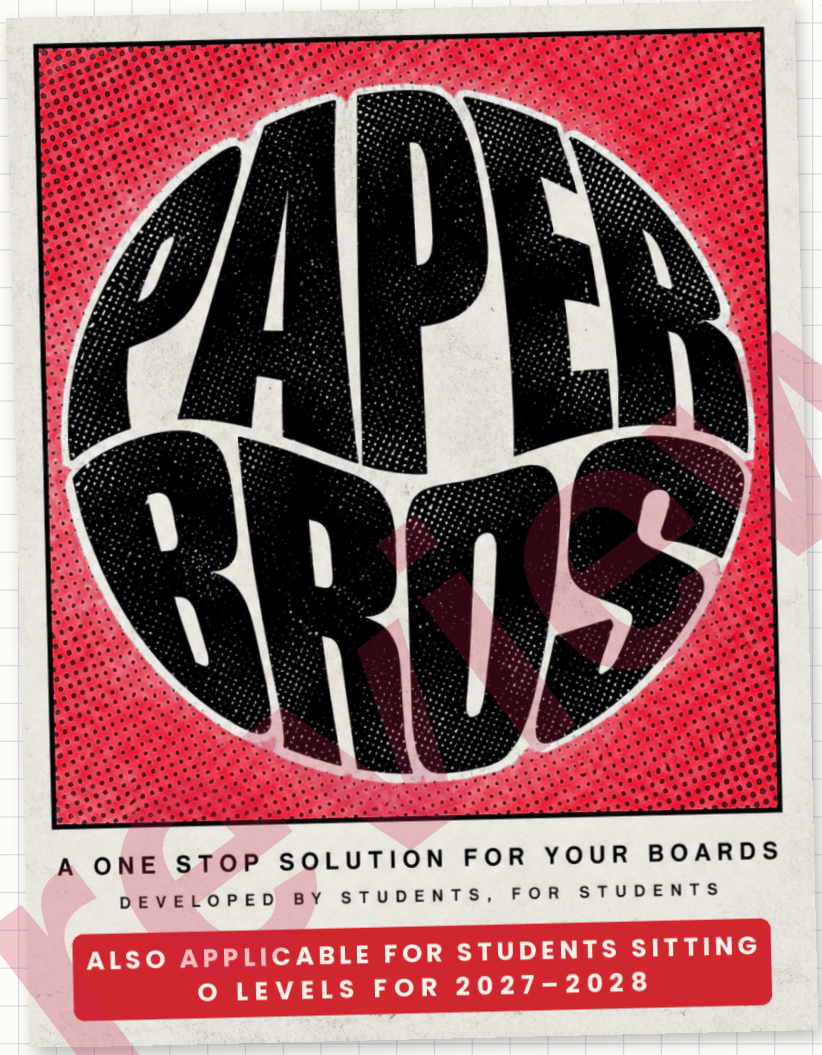
PREVIEW

Cambridge O Level

Chemistry 5070

the complete notes – for exams in 2029



notes for every syllabus point · worked examples
accurate diagrams & graphs · Paper 1 & Paper 2 practice + answers

this notebook belongs to:

Name .....

School / class ..... Exam: June / Nov

Preview

PREVIEW

what's inside

Contents

tick the box when a topic is done & dusted ✓

#	Topic	What's inside (syllabus sections)	Notes pg	Practice pg	✓
–	The exam at a glance	papers, marks, AOs, command words, how to use these notes	3	–	☐
–	Toolkit pages	key definitions · ions, formulae & colours · how to score highly · Periodic Table · revision tracker	4	–	☐
1	States of matter	1.1 solids, liquids & gases, changes of state, heating/cooling curves, gas volume · 1.2 diffusion	10	16	☐
2	Atoms, elements & compounds	2.1 elements, compounds, mixtures · 2.2 atomic structure · 2.3 isotopes · 2.4 ionic bonding · 2.5 covalent bonding · 2.6 giant covalent · 2.7 metallic bonding	20	31	☐
3	Stoichiometry	3.1 formulas & equations · 3.2 A_r and M_r · 3.3 the mole, gas volumes, concentration, titration, empirical formula, % yield & purity	35	44	☐
4	Electrochemistry	4.1 electrolysis, half-equations, electroplating · 4.2 hydrogen–oxygen fuel cells	48	56	☐
5	Chemical energetics	5.1 exo/endothermic, ΔH, activation energy, pathway diagrams, bond energy calculations	60	66	☐
6	Chemical reactions	6.1 physical/chemical change · 6.2 rate of reaction · 6.3 reversible reactions, equilibrium, Haber & Contact · 6.4 redox	70	80	☐
7	Acids, bases & salts	7.1 properties of acids & bases, indicators, pH, strong/weak · 7.2 oxides · 7.3 preparing salts, solubility rules	84	93	☐
8	The Periodic Table	8.1 arrangement & trends · 8.2 Group I · 8.3 Group VII · 8.4 transition elements · 8.5 noble gases	97	105	☐
9	Metals	9.1 properties · 9.2 uses · 9.3 alloys · 9.4 reactivity series · 9.5 corrosion · 9.6 extraction (iron & aluminium)	109	119	☐
10	Chemistry of the environment	10.1 water · 10.2 fertilisers · 10.3 air quality & climate	124	132	☐
11	Organic chemistry	11.1 formulas & terms · 11.2 naming · 11.3 fuels · 11.4 alkanes · 11.5 alkenes · 11.6 alcohols · 11.7 carboxylic acids · 11.8 polymers	136	148	☐
12	Experimental techniques & chemical analysis	12.1 apparatus & terms · 12.2 titrations · 12.3 chromatography · 12.4 separation & purification · 12.5 tests for ions & gases · practical-paper skills	153	162	☐
P4	Alternative to Practical	reading scales, planning, graphs, evaluation + Paper 4 practice	167	170	☐
★	Mock examination	full Paper 1 (40 MCQ), Paper 2 (80 marks) and Paper 4 (40 marks) under timed conditions	–	177	☐
⚡	Last-minute cram	equation bank, calculation bank, observation bank, topics in 60 seconds, 60 quick-fire questions	184	–	☐
✶	Answers	Paper 1 keys + Paper 2 mark-scheme style answers for every topic	192	–	☐

Same content as 2027 & 2028 ✔

Cambridge says the 2029 syllabus has **no significant changes which affect teaching** – so these notes work for the June/Nov 2027 and 2028 series too. Textbooks endorsed from 2023 are still fine.

How I'd use this book

1. read the notes → 2. cover them & redo the worked examples → 3. do the practice without peeking → 4. mark with the answers at the back & fix your mistakes in red.

Preview

PREVIEW

before we start...

The Exam at a Glance

You sit **three** papers. Everyone takes Paper 1 **and** Paper 2, then **either** Paper 3 **or** Paper 4. Grades go from **A*** down to **E**.

Paper	Time	Marks	Weight	What it looks like	Tests
1 Multiple Choice	1 h	40	30%	40 compulsory four-option (A–D) questions	AO1 + AO2
2 Theory	1 h 45 min	80	50%	compulsory short-answer & structured questions	AO1 + AO2
3 Practical Test	1 h 30 min	40	20%	you actually do experiments in a lab	AO3
			– OR –		
4 Alternative to Practical	1 h	40	20%	written questions about experiments (no lab)	AO3

AO Assessment objectives

Objective	Weight
AO1 Knowledge with understanding – recall facts, definitions, equations, safety, uses	50%
AO2 Handling information & problem-solving – data, graphs, calculations, unfamiliar contexts	30%
AO3 Experimental skills & investigations – planning, measuring, evaluating	20%

Inside Papers 1 & 2 the split is roughly **63% AO1 : 37% AO2** – so about a third of the marks are "apply it to something new". Don't just memorise – practise!

✱ You are GIVEN in the exam

- a **Periodic Table** (with A_r values) – use it, e.g. Cl = 35.5
- molar gas volume: 24 dm^3 at r.t.p.
- **Notes for qualitative analysis** (ion, gas & flame tests) in Papers 3 & 4 – but Paper 1 & 2 can still ask them, so learn them (Topic 12)
- a calculator is allowed in all papers

key How to read these notes

- **maroon bold** = a key word that earns the mark
- yellow highlight = learn word-for-word
- pink / blue / green = colour-coded comparisons
- yellow box = **definition**; blue box = **worked example**
- sticky notes = exam tips & traps ⚠
- little grey numbers like [2.4.3] = the syllabus point

cmd Command words

The first word of the question tells you what the examiner wants. These are the ones that catch people out:

Word	What you do
State / Give	short clear answer, no explanation needed
Define	the exact meaning (learn definitions word-for-word)
Describe	say what happens / the main features (observations!)
Explain	give the reasons *why* or *how* – use "because..." and particle ideas
Suggest	apply what you know to a new situation; several answers may be right
Deduce	work it out from the information given
Predict	say what will happen using a pattern or data
Calculate	work out a number – show working + units
Determine	get an answer using the data/graph given
Compare	similarities and/or differences – mention both things
Contrast	differences only
Identify	name / pick out / recognise
Evaluate	judge the quality / value of something (pros + cons)
Outline	main points only, briefly
Justify	back your answer with evidence
Sketch	simple drawing showing key features (correct shape)
Show (that)	structured working that ends at the given answer

Top 3 mark-losers

① no **units** or wrong sig. figs ② "it reacts" instead of an **observation** (fizzing, colour change, precipitate) ③ explaining with "atoms" when it's **ions** or **molecules**.

Preview

PREVIEW

learn these word-for-word

Key Definitions

Definition questions are the **easiest marks** in the exam — but only if you use the **key words** (in bold). Cover the right column and test yourself: read one aloud, cover it, write it from memory, check the bold words. Five a day and you'll know them all in two weeks.

term	definition
Topics 1–2 · particles & bonding	
diffusion	the **net movement** of particles from a region of **higher** to **lower concentration** as a result of their **random motion**
element	a substance made of only **one type of atom**
compound	two or more different elements **chemically combined** in a **fixed ratio**
mixture	two or more substances **not chemically combined**
proton (atomic) number	the number of **protons** in the nucleus of an atom
mass (nucleon) number	the total number of **protons and neutrons** in the nucleus
isotopes	different atoms of the **same element** with the same number of **protons** but different numbers of **neutrons**
ion	an atom or group of atoms that has **gained or lost electrons** and so carries a charge
ionic bond	a strong **electrostatic attraction** between **oppositely charged ions**
covalent bond	a **shared pair of electrons** between two atoms, giving **noble gas** configurations
metallic bonding	electrostatic attraction between **positive ions** in a giant lattice and a **'sea' of delocalised electrons**
Topic 3 · stoichiometry	
molecular formula	the number and type of different **atoms in one molecule**
empirical formula	the **simplest whole number ratio** of the different atoms or ions in a compound
relative atomic mass, A_r	the **average mass** of the isotopes of an element compared with $^1/_{12}$ of the mass of an atom of **carbon-12**
relative molecular mass, M_r	the **sum** of the relative atomic masses of all the atoms in the formula (relative formula mass for ionic compounds)
mole	the unit of amount of substance; one mole contains **6.02×10^{23}** particles (the Avogadro constant)
molar gas volume	**24 dm^3** per mole for any gas at room temperature and pressure (r.t.p.)
concentration	the amount of solute in a given volume of solution, in **mol/dm^3** or **g/dm^3**
limiting reactant	the reactant that is **used up first**; it decides the amount of product

term	definition
Topic 5 · energetics	
exothermic	a reaction that **transfers thermal energy to the surroundings**, so the temperature of the surroundings **increases**
endothermic	a reaction that **takes in thermal energy from the surroundings**, so the temperature of the surroundings **decreases**
enthalpy change, ΔH	the **transfer of thermal energy** during a reaction; negative for exothermic, positive for endothermic
activation energy, E_a	the **minimum energy** that colliding particles must have to **react**
Topic 6 · reactions	
catalyst	a substance that **increases the rate** of a reaction and is **unchanged** at the end
dynamic equilibrium	in a **closed system**, the rate of the forward reaction **equals** the rate of the reverse reaction and the **concentrations** no longer change
oxidation	**gain of oxygen**, **loss of electrons**, or an **increase in oxidation number**
reduction	**loss of oxygen**, **gain of electrons**, or a **decrease in oxidation number**
redox	a reaction in which oxidation and reduction happen **at the same time**
oxidising agent	a substance that **oxidises** another substance and is itself **reduced**
reducing agent	a substance that **reduces** another substance and is itself **oxidised**
Topic 7 · acids, bases & salts	
acid	a **proton (H^+) donor**
base	a **proton (H^+) acceptor**
alkali	a **soluble base**; it produces **OH^-** ions in water
strong acid	an acid that is **completely dissociated** (ionised) in aqueous solution
weak acid	an acid that is **partially dissociated** in aqueous solution
neutralisation	the reaction of H^+ and OH^- to form **water**: $H^+ + OH^- \longrightarrow H_2O$
amphoteric oxide	an oxide that reacts with **both acids and bases** (e.g. Al_2O_3, ZnO)
salt	a compound formed when the **hydrogen** of an acid is **replaced** by a metal or ammonium ion
hydrated / anhydrous	**chemically combined with water** / containing **no water**
water of crystallisation	water molecules **present in the crystals** of a hydrated salt

Preview

PREVIEW

term	definition
	Topic 4 · electrochemistry
electrolysis	the **decomposition** of an ionic compound, when **molten or in aqueous solution**, by the passage of an **electric current**
electrolyte	the **molten or aqueous** substance that undergoes electrolysis
anode / cathode	the **positive** electrode / the **negative** electrode
fuel cell	uses hydrogen and oxygen to produce **electricity** with **water as the only product**
	Topics 8–9 · Periodic Table & metals
group / period	a **vertical column** / a **horizontal row** of the Periodic Table
alloy	a **mixture** of a metal with other elements (e.g. brass: Cu + Zn)
ore	a rock containing enough of a metal compound to make extraction **worthwhile**
rusting	the corrosion of iron and steel; needs both **water and oxygen** to form hydrated iron(III) oxide
galvanising	coating iron or steel with **zinc**, which acts as a barrier and as **sacrificial** protection
sacrificial protection	a **more reactive** metal is attached and corrodes **instead of** the iron
slag	calcium silicate formed in the blast furnace when CaO removes the **acidic impurity** SiO_2
	Topic 10 · environment
greenhouse effect	greenhouse gases **absorb** infrared radiation from the Earth's surface and **re-emit** it, warming the atmosphere
climate change	long-term changes in weather patterns, caused by the increase in **average global temperature**
eutrophication	excess **nitrates/phosphates** in water ⟶ algal growth ⟶ plants die ⟶ bacteria decompose them using up **dissolved oxygen** ⟶ aquatic life dies
NPK fertiliser	supplies the elements **nitrogen, phosphorus and potassium** for improved plant growth
acid rain	rain with a pH below about 5, caused by dissolved **SO_2 and NO_x**
	Topic 12 · practical vocabulary
solvent / solute	the liquid that **dissolves** / the substance that **is dissolved**
solution	a mixture of one or more **solutes dissolved in a solvent**
saturated solution	a solution that contains the **maximum** amount of solute that can dissolve at that temperature
residue / filtrate	the solid left **in the filter paper** / the liquid that **passes through**
pure substance	a **single** substance with nothing else mixed with it; sharp m.p. and b.p.
R_f value	distance travelled by the **substance** ÷ distance travelled by the **solvent front**
locating agent	a chemical sprayed on a chromatogram to make **colourless** spots visible
titre	the volume of solution added from the burette to reach the **end-point**

term	definition
	Topic 11 · organic chemistry
hydrocarbon	a compound containing **hydrogen and carbon only**
homologous series	a family of compounds with the **same general formula**, the **same functional group**, similar chemical properties and a **trend** in physical properties; each member differs by CH_2
functional group	an atom or group of atoms that **determines the chemical properties** of a homologous series
structural isomers	compounds with the **same molecular formula** but **different structural formulae**
saturated	contains only **single** carbon–carbon bonds
unsaturated	contains one or more carbon–carbon **double** (or triple) bonds
fraction	a **mixture** of hydrocarbons with **similar boiling points** from fractional distillation
cracking	the **thermal decomposition** of long-chain alkanes into shorter alkanes and alkenes (high temperature + catalyst)
substitution	a reaction in which one atom or group is **replaced** by another (alkane + Cl_2, UV)
addition reaction	two molecules **join** to make one product (across a C=C double bond)
polymer	a **large molecule** built up from many **small molecules** called monomers
addition polymerisation	monomers with a C=C bond join; the polymer is the **only product**
condensation polymerisation	monomers join and a **small molecule** (e.g. water) is **eliminated** at each link
fermentation	the breakdown of glucose by **yeast** in the **absence of oxygen** to make ethanol and CO_2
esterification	a carboxylic acid + an alcohol ⟶ an **ester + water** (acid catalyst)

Self-test ✎ cover the page and define:

- ☐ isotopes
- ☐ empirical formula
- ☐ electrolysis
- ☐ activation energy
- ☐ catalyst
- ☐ weak acid
- ☐ oxidising agent
- ☐ homologous series
- ☐ structural isomers
- ☐ eutrophication

examiners' favourite 🎯

"**Define**" questions are usually **1–2 marks**. If two key ideas are in the definition, there are two marks — leave one out and you lose one.

Preview

PREVIEW

keep this page open while you revise

Ions, Formulae & Colours Toolkit

Ions you must know

charge	positive ions	negative ions
1	H^+ Li^+ Na^+ K^+ Ag^+ **NH_4^+**	F^- Cl^- Br^- I^- **OH^-** **NO_3^-**
2	Mg^{2+} Ca^{2+} Ba^{2+} Cu^{2+} Zn^{2+} Fe^{2+} Pb^{2+}	O^{2-} S^{2-} **SO_4^{2-}** **SO_3^{2-}** **CO_3^{2-}**
3	Al^{3+} Fe^{3+} Cr^{3+}	N^{3-} **PO_4^{3-}**

bold = compound ions: keep them together and use **brackets** if you need more than one: $Ca(OH)_2$, $(NH_4)_2SO_4$.

Formulae used again and again

acids	alkalis / bases	gases
HCl hydrochloric	NaOH sodium hydroxide	H_2 O_2 N_2 Cl_2
H_2SO_4 sulfuric	KOH potassium hydroxide	CO_2 CO SO_2
HNO_3 nitric	$NH_3(aq)$ ammonia	NH_3 CH_4 NO_2
CH_3COOH ethanoic	$Ca(OH)_2$ limewater	$H_2O(g)$ steam

Diatomic elements: H_2 N_2 O_2 F_2 Cl_2 Br_2 I_2 – "Have No Fear Of Ice Cold Beverages".

Key numbers

Avogadro constant	6.02×10^{23} /mol
molar gas volume (r.t.p.)	24 dm^3 = 24 000 cm^3
r.t.p.	20 °C and 1 atm
1 dm^3	1000 cm^3 (= 1 litre)
pure water	m.p. 0 °C, b.p. 100 °C, pH 7

Tests in one box

for	add...	positive result
CO_3^{2-}	dilute acid	fizzes; limewater milky
SO_4^{2-}	HNO_3 then $Ba(NO_3)_2$	white ppt.
Cl^-/Br^-/I^-	HNO_3 then $AgNO_3$	white / cream / yellow ppt.
NO_3^-	NaOH + Al foil, warm	NH_3: red litmus ⟶ blue
water	anhydrous $CuSO_4$	white ⟶ blue
C=C	bromine water	orange ⟶ colourless

Write the formula

calcium nitrate · aluminium sulfate · ammonium carbonate · iron(II) hydroxide · copper(II) chloride · sodium phosphate

$Ca(NO_3)_2$ | $Al_2(SO_4)_3$ | $(NH_4)_2CO_3$ | $Fe(OH)_2$ | $CuCl_2$ | Na_3PO_4

Reactivity series

K > Na > Ca > Mg > Al > (C) > Zn > Fe > Pb > (H) > Cu > Ag > Au

"Please Send Charlie's Monkeys And Cute Zebras Into Lovely Hot Countries Signed Gordon"

The colour cheat-sheet

substance	colour
solids	
CuO, MnO_2, carbon (soot)	black
$CuSO_4 \cdot 5H_2O$ / anhydrous $CuSO_4$	blue / white
$CoCl_2$ hydrated / anhydrous	pink / blue
copper metal	pink-brown
iodine / sulfur	grey-black / yellow
ZnO	yellow when hot, white when cold
Fe_2O_3 (rust)	red-brown
solutions	
$Cu^{2+}(aq)$ / $Fe^{2+}(aq)$ / $Fe^{3+}(aq)$	blue / pale green / yellow-brown
MnO_4^- / $Cr_2O_7^{2-}$	purple / orange
$Br_2(aq)$ / $I_2(aq)$	orange / brown
precipitates	
$Cu(OH)_2$ / $Fe(OH)_2$ / $Fe(OH)_3$	blue / green / red-brown
$Al(OH)_3$, $Zn(OH)_2$, $Ca(OH)_2$, $BaSO_4$, AgCl	white
AgBr / AgI / PbI_2	cream / yellow / bright yellow
gases	
Cl_2 / NO_2 / Br_2 vapour / I_2 vapour	yellow-green / brown / red-brown / purple
flames	
Li^+ / Na^+ / K^+	red / yellow / lilac
Ca^{2+} / Ba^{2+} / Cu^{2+}	orange-red / light green / blue-green

colour = free marks

Around **5–8 marks** per exam come from simply knowing colours. Make flashcards from this table!

Solubility in one line

Always soluble: Na^+, K^+, NH_4^+ salts and all nitrates. **Insoluble:** AgCl, $PbCl_2$, $BaSO_4$, $PbSO_4$, ($CaSO_4$ slightly), most carbonates and hydroxides.

Preview

PREVIEW

work smarter, not just harder

How to Score Highly

The 7 golden rules for Paper 2

1. **Circle the command word** (state / describe / explain / suggest) and answer *that* question.
2. **Marks = points.** [3] means three separate, creditworthy ideas.
3. **Use the data** given: quote numbers from tables and graphs.
4. **Equations:** correct formulas first, then balance, then state symbols if asked.
5. **Calculations:** show every step, give **units** and **3 s.f.**
6. **Observations** = what you see/hear/smell (fizzing, colour change, precipitate), never "a gas is produced".
7. Use **key words**: "electrostatic attraction", "delocalised electrons", "frequency of collisions", "intermolecular forces".

Paper 1 (multiple choice) tips

- about **90 seconds** per question — don't get stuck
- cover the options, answer in your head, then look
- eliminate the clearly wrong ones; two are usually close
- watch for **NOT** and **LEAST** in the stem
- never leave a blank: there is no negative marking

same topic, different command words

State the trend in reactivity of Group 1. ⟶ "increases down the group."

Explain the trend. ⟶ "the outer electron is further from the nucleus, so it is less strongly attracted and lost more easily."

Predict rubidium + water. ⟶ "reacts more vigorously than potassium, forming RbOH and H_2."

An 8-week plan using this book

week	do this
1	Topics 1–2: notes + extra pages; definitions page
2	Topic 3: every worked example, then the mole gym
3	Topics 4–5: half-equations + bond energy calcs
4	Topics 6–7: rates, equilibrium, salts
5	Topics 8–10: tables of trends, pollution
6	Topics 11–12: organic master table, ion tests
7	Alternative to Practical section + all "on one page" maps
8	**Mock exam** under timed conditions; mark it; redo weak topics; cram pages

Each study session (45 min)

1. 10 min: read the notes, highlight what's new
2. 15 min: cover the worked examples and redo them
3. 15 min: practice questions — mark with the answers section
4. 5 min: tick the "I can…" list on the topic map

On the day ✔

- two black pens, pencil, ruler, eraser, **calculator**
- use the **Periodic Table** at the back of the paper for every A_r
- check the back page — there may be one more question!
- last 10 minutes: check units, signs (ΔH) and state symbols

the 2-minute rule ⏱

Stuck on a part of a question for more than two minutes? Mark it with a star, move on, and come back at the end. Fresh eyes find the answer.

Top 10 ways students throw marks away

✕ the slip	✓ the fix
"the bonds break" when a simple molecular substance melts	"weak **intermolecular forces** are overcome"
"electrons move" in molten/aqueous ionic compounds	"**ions** are free to move"
"more collisions"	"more **frequent** collisions / more collisions **per second**"
chlorine written as Cl in an equation	**Cl_2** (diatomic)
volume left in cm^3 in n = c × V	convert to **dm^3** first
no sign on ΔH	− for exothermic, + for endothermic
"goes clear" for bromine water	"**colourless**"
"carbon dioxide is given off" as an observation	"**effervescence / fizzing**"
A_r of Cl taken as 35 or 36	use **35.5** from the Periodic Table
rounding during a calculation	keep all digits; round only at the **end** (3 s.f.)

Calculation checklist

☐ equation balanced ☐ moles first ☐ ratio used ☐ units converted ☐ final unit written ☐ 3 s.f. ☐ answer sensible?

use the free data the exam paper gives you a **Periodic Table** (A_r values, proton numbers, symbols) and the molar gas volume. Never guess a relative atomic mass.

Preview

data page · use it for every calculation

The Periodic Table

	I	II										III	IV	V	VI	VII	VIII
1	1 H 1																2 He 4
2	3 Li 7	4 Be 9										5 B 11	6 C 12	7 N 14	8 O 16	9 F 19	10 Ne 20
3	11 Na 23	12 Mg 24										13 Al 27	14 Si 28	15 P 31	16 S 32	17 Cl 35.5	18 Ar 40
4	19 K 39	20 Ca 40	21 Sc 45	22 Ti 48	23 V 51	24 Cr 52	25 Mn 55	26 Fe 56	27 Co 59	28 Ni 59	29 Cu 64	30 Zn 65	31 Ga 70	32 Ge 73	33 As 75	34 Se 79	35 Br 80	36 Kr 84
5	37 Rb 85	38 Sr 88	39 Y 89	40 Zr 91	41 Nb 93	42 Mo 96	43 Tc -	44 Ru 101	45 Rh 103	46 Pd 106	47 Ag 108	48 Cd 112	49 In 115	50 Sn 119	51 Sb 122	52 Te 128	53 I 127	54 Xe 131
6	55 Cs 133	56 Ba 137	57 La 139	72 Hf 178	73 Ta 181	74 W 184	75 Re 186	76 Os 190	77 Ir 192	78 Pt 195	79 Au 197	80 Hg 201	81 Tl 204	82 Pb 207	83 Bi 209	84 Po -	85 At -	86 Rn -

Z = proton (atomic) number
X = symbol
A = relative atomic mass, A_r

La = lanthanoids (57–71) · period 7 (Fr, Ra, actinoids) not shown

✱ Reading a box

- small number (top) = **proton number** = number of protons = electrons in the atom
- bottom number = A_r (use it for M_r and moles)
- neutrons in the most common isotope ≈ $A_r - Z$ (rounded)
- group (I–VIII) = outer electrons; period (row) = number of shells

colour	family
	Group I: alkali metals
	Group II
	transition elements
	other metals
	metalloids (on the staircase)
	non-metals
	Group VII: halogens
	Group VIII: noble gases

WE ✎ use the table

Find M_r of magnesium nitrate, $Mg(NO_3)_2$.
Mg 24 + 2 × (N 14 + 3 × O 16) = 24 + 2 × 62 = **148**

WE ✎ predict from position

Strontium (Sr, Z = 38) is in Group II, Period 5. Predict its ion and the formula of its chloride.
2 outer electrons ⇒ Sr^{2+} ⇒ chloride $SrCl_2$; more reactive than calcium (further down).

exam tip the real exam paper prints this table on its last page with the A_r values. Practise finding elements fast — it saves minutes.

molar volume one mole of any gas occupies **24 dm³** at room temperature and pressure (r.t.p.).

✱ Symbols that trip people up

Na sodium (natrium)	**K** potassium (kalium)	**Fe** iron (ferrum)	**Cu** copper (cuprum)	**Ag** silver (argentum)
Au gold (aurum)	**Pb** lead (plumbum)	**Sn** tin (stannum)	**Hg** mercury	**W** tungsten (wolfram)

careful a capital letter starts every symbol, the second letter is **lower case**: **Co** is cobalt, but **CO** is carbon monoxide!

Preview

PREVIEW

see your progress grow

My Revision Tracker

Fill this in as you go. Confidence: 1 = "help!", 5 = "I could teach it". Revisit anything below 4 the week before the exam.

#	topic	notes read	map ticked	practice score	challenge 1	challenge 2	confidence 1–5
1	States of matter	☐	☐	/	/	/	
2	Atoms, elements & compounds	☐	☐	/	/	/	
3	Stoichiometry	☐	☐	/	/	/	
4	Electrochemistry	☐	☐	/	/	/	
5	Chemical energetics	☐	☐	/	/	/	
6	Chemical reactions	☐	☐	/	/	/	
7	Acids, bases & salts	☐	☐	/	/	/	
8	The Periodic Table	☐	☐	/	/	/	
9	Metals	☐	☐	/	/	/	
10	Chemistry of the environment	☐	☐	/	/	/	
11	Organic chemistry	☐	☐	/	/	/	
12	Experimental techniques	☐	☐	/	/	/	
P4	Alternative to Practical	☐	-	/ 68	-	-	

also done...	✓	also done...	✓
key definitions (all)	☐	equation bank written from memory	☐
colour cheat-sheet flashcards	☐	quick-fire 60 questions	☐
Alternative to Practical notes	☐	mock Paper 1 marked	☐
all 12 "on one page" maps	☐	mock Paper 2 marked	☐

✱ Mock exam log

date	Paper 1 / 40	Paper 2 / 80	%

✱ My weak spots to fix

✱ Definitions mastered

Colour one square for every definition (from page 4) you can write perfectly:

☐☐☐☐☐☐☐☐☐☐
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small wins 🏆

Ticking boxes isn't childish — seeing progress keeps you going. Reward yourself after every topic you finish!

Preview

PREVIEW

Topic 1

States of Matter

solids, liquids, gases & what their particles get up to

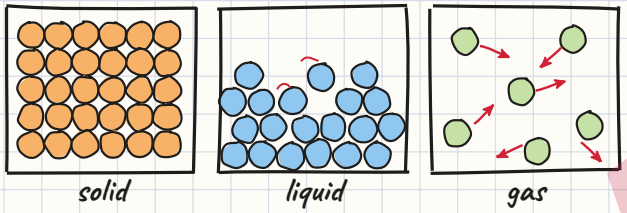
1.1 Solids, liquids & gases

✱ What you can actually see [1.1.1]

	solid	liquid	gas
shape	fixed	takes shape of container	fills the whole container
volume	fixed	fixed	not fixed
flows?	no	yes	yes
compress?	no	hardly	easily
density	high	fairly high	very low

✱ The particle model (kinetic particle theory) [1.1.2]

All matter is made of tiny **particles** (atoms, ions or molecules) that are **always moving**. How much they move depends on how much **kinetic energy** they have.

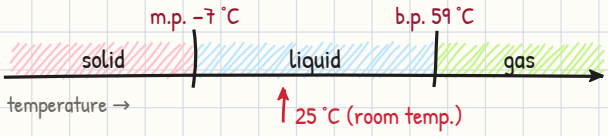


	solid	liquid	gas
separation	very close (touching)	close (mostly touching)	far apart
arrangement	regular (lattice)	irregular / random	random
motion	vibrate about fixed positions	move around, slide past each other	move fast, randomly, in all directions
attractive forces	strong	weaker	very weak
energy	lowest	more	highest

why? Gases compress easily because there is **lots of empty space** between particles. Solids & liquids can't – particles are already touching.

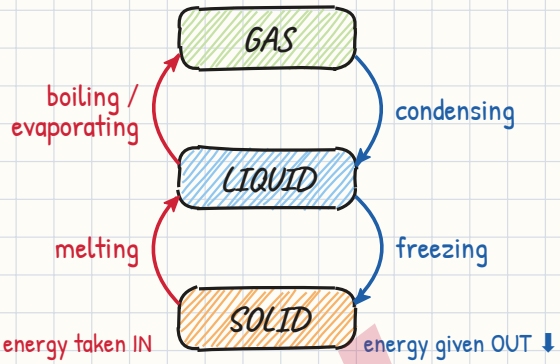
✱ Using m.p. & b.p. data to find the state

At a temperature T: T below m.p. ⟶ solid between m.p. and b.p. ⟶ liquid above b.p. ⟶ gas



e.g. bromine: m.p. −7 °C, b.p. 59 °C ⇒ at 25 °C it is a **liquid** (25 is between the two). At 70 °C it would be a gas.

✱ Changes of state [1.1.3]



- **Melting** (at the **melting point**): heating gives particles more kinetic energy ⟶ they **vibrate more** ⟶ they overcome the forces of attraction holding them in the lattice ⟶ they can move past each other.
- **Boiling** (at the **boiling point**): particles *throughout* the liquid gain enough energy to overcome the attractive forces and escape as a gas ⟶ bubbles form inside the liquid.
- **Evaporating**: happens at **any temperature below the b.p.**, only at the **surface**. The most energetic particles escape, so the liquid left behind **cools down** (that's why sweat cools you!).
- **Condensing**: a gas cools ⟶ particles lose energy, slow down ⟶ attractive forces pull them close together ⟶ liquid.
- **Freezing**: a liquid cools ⟶ particles lose energy ⟶ settle into a fixed, regular lattice. freezing point = melting point for a pure substance.

	evaporation	boiling
temperature	any temp. below b.p.	only at the b.p.
where?	surface only	throughout the liquid (bubbles)
speed	slow	fast

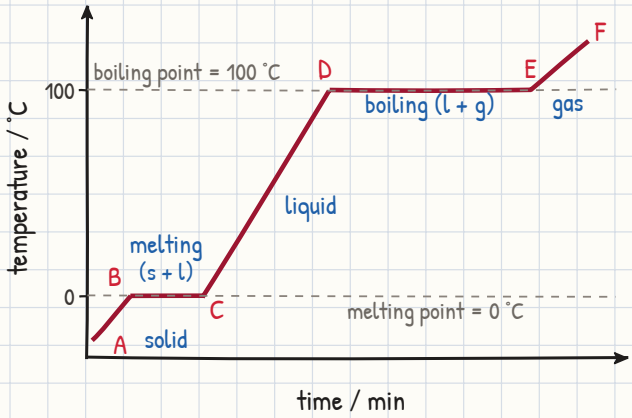
exam tip ✎

In a "describe/explain" particle question always cover 3 **things**: arrangement, separation and motion – plus energy & forces if it's a change of state. One mark each, usually!

Preview

1.1 Heating & cooling curves

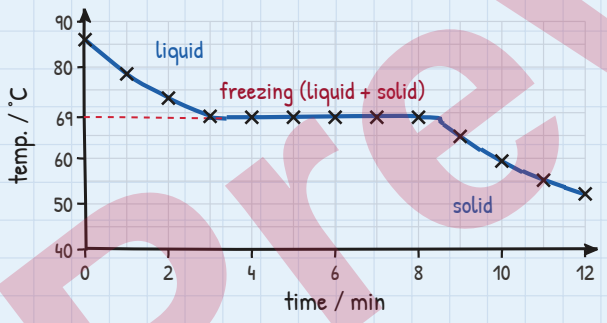
[1.1.4] – e.g. heating ice at a steady rate until it's all steam



- **Sloping parts** (AB, CD, EF): energy goes into increasing the particles' **kinetic energy** ⟶ temperature rises.
- **Flat parts** (BC, DE): temperature stays constant because the energy taken in is used to overcome the forces of attraction between particles, not to speed them up.
- DE is **longer** than BC: much more energy is needed to separate particles completely (liquid ⟶ gas) than just loosen them (solid ⟶ liquid).
- A **pure** substance gives a **sharp** flat line. An impure one melts over a **range** (and melts lower / boils higher).

Worked example ✎ cooling curve

Liquid X (molten wax) cools in a boiling tube; temperature taken every minute:

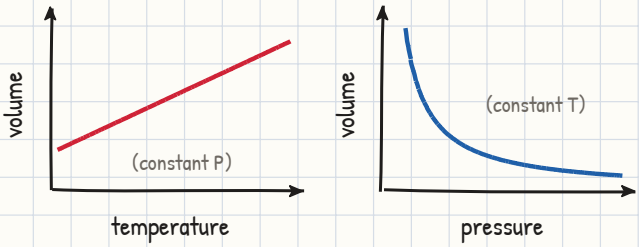


Q: What is the freezing point of X? Why is the line flat from 3 to 8 min?

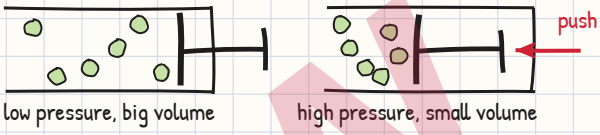
A: 69 °C – read off the flat part. While freezing, energy is **released** as attractive forces form between particles; this balances the heat lost to the surroundings, so the temperature stays constant until all of X is solid.

✱ Gases: temperature, pressure & volume [1.1.5]

Gas **pressure** = gas particles **colliding with the walls** of the container.



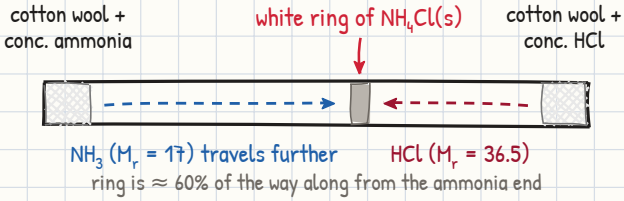
- **Temperature ↑** (pressure kept constant): particles gain kinetic energy, **move faster** and hit the walls harder and more often ⟶ the gas expands, volume increases.
- **Pressure ↑** (temperature kept constant): the particles are pushed **closer together** ⟶ volume decreases. Double the pressure ≈ half the volume.



1.2 Diffusion

Diffusion: the net movement of particles from a region of higher concentration to a region of lower concentration, caused by the **random motion** of the particles. [1.2.1]

- Fast in **gases**, slower in liquids (particles move slower, less space), basically none in solids.
- **Higher temperature** ⟶ particles move faster ⟶ faster diffusion.
- Lower M_r ⟶ faster diffusion. At the same temperature lighter molecules move faster. [1.2.2]



$$NH_3(g) + HCl(g) \longrightarrow NH_4Cl(s)$$

Worked example ✎

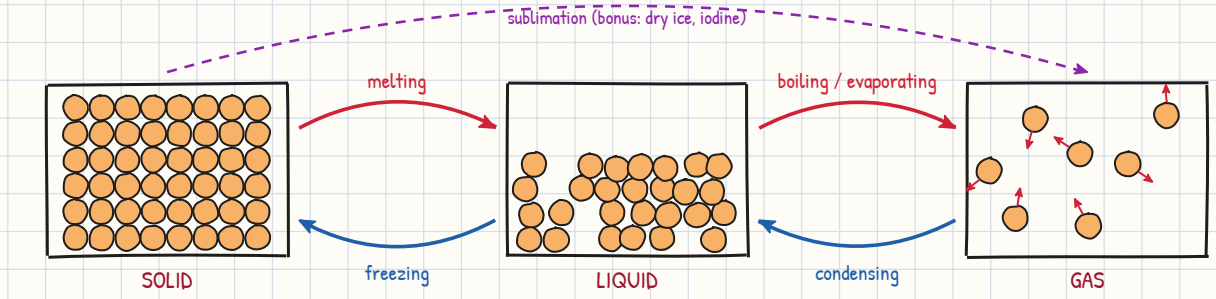
Q: Explain why the white ring forms nearer the HCl end. [3]

A: ① both gases **diffuse** along the tube by random motion; ② NH_3 has a **lower M_r** (17) than HCl (36.5), so its molecules **move faster**; ③ NH_3 travels **further** in the same time, so they meet closer to the HCl end.

Preview

1.1 Kinetic particle theory, in depth

Everything on this page comes from one idea: **all matter is made of tiny particles that are always moving.** The hotter something is, the more **kinetic energy** its particles have. Changes of state happen when particles gain enough energy to overcome the **forces of attraction** holding them together — or lose enough energy for those forces to pull them back together. [1.1.2–3]



red = energy taken IN (endothermic)

blue = energy given OUT (exothermic)

change	what happens to the particles
melting	particles gain energy, **vibrate more** and partly overcome the forces holding them in the lattice; they can now **slide past** each other
boiling	particles **throughout** the liquid gain enough energy to **completely overcome** the forces; bubbles of gas form inside the liquid
evaporating	only the **most energetic** particles **at the surface** escape; happens at any temperature below the b.p.
condensing	gas particles lose energy, slow down and **attractive forces** pull them close together into a liquid
freezing	liquid particles lose energy and settle into **fixed positions** in a regular lattice

how fast are they? 🚀

At room temperature an average oxygen molecule in the air is moving at about **480 m/s** — nearly twice the cruising speed of a jet plane! You don't get blown over because the collisions come from every direction and cancel out.

WE ✎ explain melting, 3 marks

Use the kinetic particle theory to explain what happens when ice melts.

① the particles **gain (kinetic) energy** from the surroundings; ② they **vibrate more** and overcome some of the **forces of attraction** between them; ③ they can now **move past each other** (no longer in fixed positions), so the solid becomes a liquid.

WE ✎ why does sweating cool you?

① the **most energetic** water particles at the surface of the sweat **escape** (evaporate); ② the particles left behind have a **lower average kinetic energy**, so the liquid is cooler; ③ the cooler sweat takes energy from your skin.

careful particles **never** get bigger, melt or expand themselves. Only their **speed** and the **spaces between them** change.

link temperature is a measure of the **average kinetic energy** of the particles. At the same temperature, **heavier** particles move **more slowly** — that's why M_r matters in diffusion (page after next).

✱ Spot the state from a particle diagram

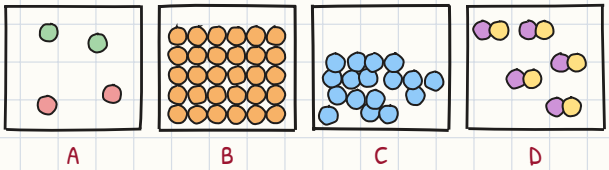
- **regular rows, all touching** ⇒ solid
- **touching but jumbled**, with a few gaps ⇒ liquid
- **far apart, random**, often with motion arrows ⇒ gas
- two different kinds of particle **not joined** ⇒ a **mixture** (Topic 2)

exam trap in a liquid the particles are still **touching**. If you draw them spaced out like a gas you lose the mark. A liquid is only about **10% less dense** than its solid; a gas is about **1000 times** less dense.

✱ Changes of state all around you

you see...	change of state
a mirror steams up after a shower	condensing (water vapour ⟶ liquid)
wet clothes dry on a washing line	evaporating
dew on the grass in the morning	condensing as the air cools at night
a puddle freezes in winter	freezing
chocolate goes soft in your pocket	melting (begins at about 30–34 °C)
bubbles all through a kettle	boiling (100 °C at sea level)

Quick check: which box shows (a) a liquid, (b) a mixture of two gases, (c) a pure gas made of molecules, (d) a solid?

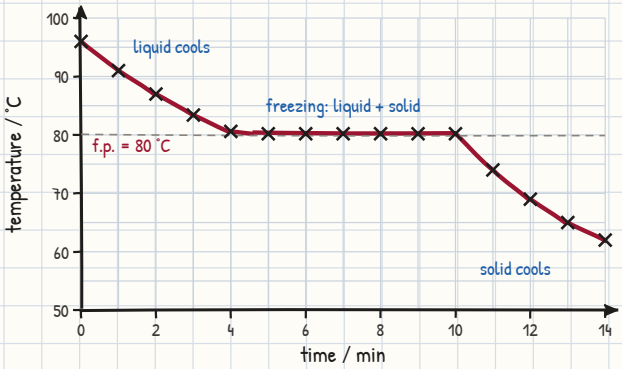


answers: (a) C (b) A (c) D (d) B

fun fact water boils at only about **70 °C** on top of Mount Everest: lower air pressure means the particles escape more easily. Pasta takes ages to cook up there!

Preview

1.1 Cooling curves & melting point data



cooling curve for naphthalene (a pure substance)

While a liquid **freezes** the temperature stays **constant**: new **forces of attraction form** as the particles settle into the lattice, and this **releases energy** that balances the heat lost to the surroundings.

✱ How to get a cooling curve (practical)

1. melt the solid in a boiling tube in a **hot water bath** (never a naked flame: it's flammable)
2. put in a **thermometer** and take the tube out of the bath
3. record the temperature **every 30 s** (stop-watch) while **stirring gently** with the thermometer, until it has been solid for a few minutes
4. plot temperature (y) against time (x); the **flat part** gives the freezing point

time / min	0	1	2	3	4	5	6
temp. / °C	96	91	87	83	81	80	80

table tip units in the **heading only**, same decimal places down a row, and read the thermometer at **eye level**.

salt on icy roads

Spreading salt on roads makes the ice **impure**. Impure water freezes below 0 °C (down to about −10 °C), so the ice melts even on a cold night. Same idea: impurities **lower** the melting point.

✱ Melting point & boiling point data questions [1.1.1]

substance	m.p. / °C	b.p. / °C	state at 25 °C
oxygen	−218	−183	gas
ethanol	−114	78	liquid
mercury	−39	357	liquid
water	0	100	liquid
sulfur	115	445	solid
sodium chloride	801	1413	solid

WE ✎ use the table

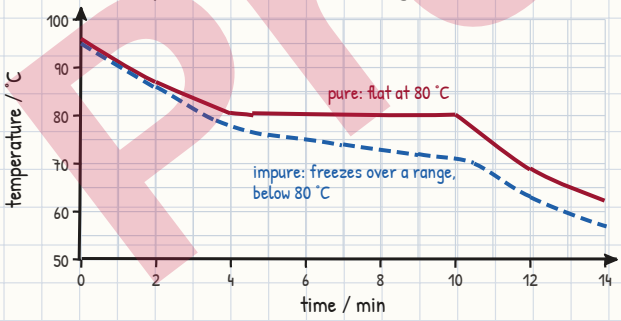
(a) state of ethanol at 100 °C? 100 > b.p. 78 ⇒ **gas**

(b) liquid over the widest range? mercury: 357 − (−39) = **396 °C** (NaCl: 612 °C but it is not liquid at room temp. — read the question!)

(c) sulfur at 300 °C? between m.p. and b.p. ⇒ **liquid**

the rule below m.p. = **solid**; between m.p. and b.p. = **liquid**; above b.p. = **gas**. Watch the minus signs: −183 is **lower** than −114.

✱ Pure or impure? Read the cooling curve



- a **pure** substance freezes (and melts) at **one sharp temperature**: a flat line
- an **impure** sample freezes **over a range** of temperatures, starting **lower**: the line slopes gently instead of going flat
- impurities also **raise** the boiling point (salty water boils above 100 °C)
- this is how chemists check the **purity** of medicines and foods (Topic 12)

WE ✎ 2 marks

A sample of naphthalene melts between 74 °C and 78 °C. Is it pure? Explain.

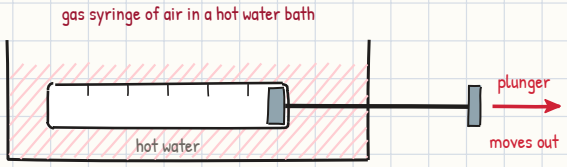
① **not pure**; ② pure naphthalene melts sharply at **80 °C**; this sample melts **below** 80 °C and **over a range**.

	heating curve	cooling curve
flat parts	melting and boiling	condensing and freezing
energy	taken in to overcome forces	released as forces form
slope	goes up	goes down

exam trap on a cooling curve, the freezing point is the temperature of the **flat part**, not where the line ends. And the plateau is **not** "when nothing happens" — it's when the most is happening!

Preview

1.1 Gases: temperature & pressure investigated



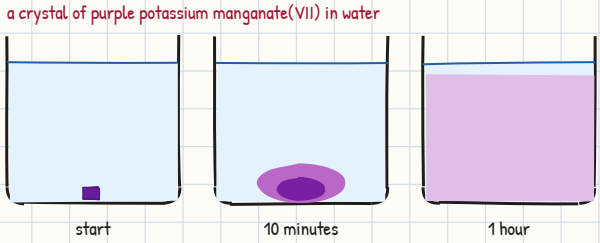
Heating a gas at constant pressure: the air particles gain kinetic energy and **move faster**; they hit the plunger **harder and more often**, pushing it out until the pressure inside equals the pressure outside. Result: the **volume increases**. [1.1.5]

Squashing a gas at constant temperature: block the end of a syringe and push the plunger in. The same particles are in a **smaller space**, so they hit the walls **more often**: the **pressure increases** (you can feel the push back).

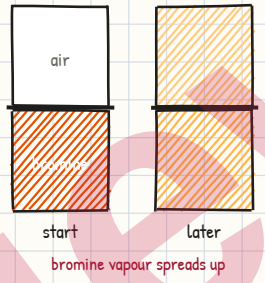
you notice...	particle explanation
a balloon shrinks in the freezer	particles slow down and hit the rubber less hard ⇒ smaller volume
tyre pressure rises on a hot day	faster particles hit the tyre wall more often and harder; volume is fixed so **pressure** rises
"do not heat" on aerosol cans	pressure rises with temperature – the can may burst
a bike pump gets harder to push	more collisions per second in a smaller volume ⇒ higher pressure

exam phrase always say **more frequent collisions** with the walls, not just "more collisions".

1.2 Diffusion experiments



The purple colour spreads through the water **without stirring**: water particles and manganate(VII) ions move randomly and the ions spread from where they are **concentrated** to where they are **not**. Diffusion in liquids is **slow** (minutes to hours); in gases it takes seconds to minutes. Solids hardly diffuse at all – their particles only vibrate.



Brown bromine vapour is **denser** than air but still spreads **upwards** into the top jar. Diffusion works in every direction – gravity cannot stop random motion.

✱ Lighter particles diffuse faster [1.2.2]

gas	H_2	He	CH_4	NH_3	O_2	HCl	CO_2	Cl_2
M_r	2	4	16	17	32	36.5	44	71

fastest ←←←←←←←← slowest

party balloons
A helium balloon goes floppy overnight, but an air-filled one lasts for days. Tiny, light helium atoms (M_r 4) move fast and escape through the rubber much more quickly than N_2 and O_2.

WE ✎ predict, 3 marks
The NH_3/HCl tube is repeated using hydrogen bromide, HBr (M_r 81), instead of HCl. Predict where the white ring forms. Explain.
① the ring forms **even closer to the HBr end**; ② HBr has a **higher M_r** (81) than HCl (36.5), so it **diffuses more slowly**; ③ NH_3 (M_r 17) travels a greater fraction of the tube before they meet.

WE ✎ temperature and diffusion
The $KMnO_4$ experiment is repeated in water at 60 °C. The colour spreads **faster**, because the particles have **more kinetic energy** and move faster.

✱ What changes the rate of diffusion?

factor	effect	why
temperature ↑	faster	particles have more kinetic energy
M_r ↓	faster	lighter particles move faster at the same temperature
state	gas > liquid > solid	more space and faster random motion in gases

Diffusion in real life: smelling food cooking from another room; a tea bag colouring the water; oxygen moving from your lungs into your blood (Biology!); the smell of a skunk spreading downwind.

exam phrase "net movement ... from a region of higher concentration to a region of lower concentration ... due to the **random motion** of particles". Learn it word for word.

Preview

PREVIEW

Topic 1 · on one page

States of Matter: the Big Picture

Solids, liquids, gases
solid: fixed shape & volume; particles touching, regular, vibrate
liquid: fixed volume, flows; close, irregular, slide past
gas: fills container; far apart, random, fast

Changes of state
melting / boiling / evaporating: energy IN
freezing / condensing: energy OUT
evaporation: surface only, any temperature

Heating & cooling curves
sloping part: KE rises, temperature rises
flat part: energy breaks forces, temperature constant
boiling plateau longer than melting plateau

STATES OF MATTER

Gas behaviour
T up (constant P): faster particles, volume up
P up (constant T): particles closer, volume down
pressure = particles hitting the walls

Diffusion
net movement high → low concentration
caused by random motion of particles
faster when hotter and when Mr is lower

Exam favourites
NH_3 / HCl tube: ring nearer the HCl end
pure substance: sharp m.p.; impure: lower, over a range
use m.p./b.p. to give the state at any temperature

I can... *tick when you can do it without notes*

- ☐ describe particle arrangement, separation and motion in all three states
- ☐ name all five changes of state and say if energy is taken in or given out
- ☐ explain why the temperature stays constant while a solid melts
- ☐ label a heating curve and say which states are present on each part
- ☐ explain the effect of temperature and pressure on gas volume
- ☐ define diffusion word-for-word and explain it with particles
- ☐ explain why NH_3 diffuses faster than HCl
- ☐ use m.p. and b.p. data to decide the state at any temperature

✱ 60-second flashcards cover the right column

question	answer
Which state can be compressed?	gas (lots of space)
Melting point of pure ice?	0 °C exactly
Energy in or out when steam condenses?	given out (exothermic)
Why is the boiling plateau longest?	all forces must be overcome to separate particles fully
Faster diffusion: CO_2 or O_2?	O_2 (lower M_r, 32 vs 44)
Evaporation vs boiling: where?	surface only vs throughout the liquid
What does an impurity do to b.p.?	raises it (and boils over a range)
Gas pressure is caused by...	particles colliding with the container walls

✱ Mistakes that cost marks ⚠

don't write...	write...
✗ "the particles expand when heated"	✓ the particles move faster and further apart; they do not change size
✗ "the temperature stays the same because nothing happens"	✓ energy is used to overcome the forces of attraction
✗ "diffusion is when particles spread out"	✓ net movement from higher to lower concentration due to random motion
✗ "gas particles stop moving when cooled"	✓ they slow down (lose kinetic energy)

Exam question decoded 🔍 3 marks

Q: Water is heated in a kettle. Explain, in terms of particles, why the temperature stays at 100 °C while the water boils, even though it is still being heated.

① the energy supplied is used to **overcome the forces of attraction** between the water particles [1]
② it is **not** used to increase the particles' **kinetic energy**, so the temperature does not rise [1]
③ the particles separate completely and escape as a **gas** (steam) [1]

examiner's note "energy breaks the bonds" is wrong: no covalent bonds break when water boils — only the forces **between** molecules.

Preview